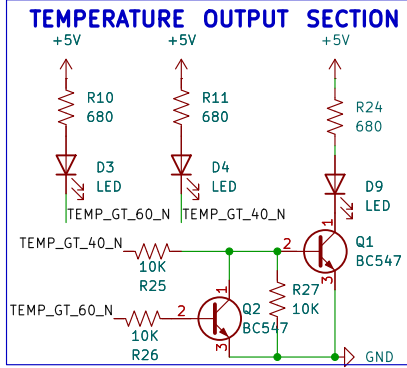
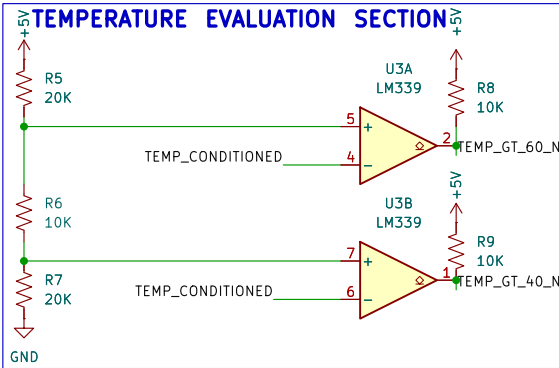
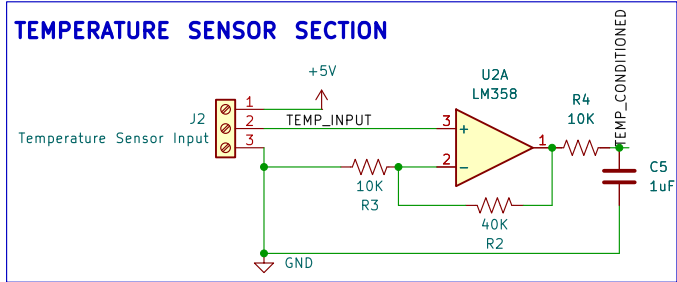


POWER SECTION

The power section circuit includes a +12V input (J1) connected to a PWR_FLAG LED (D1, 1N5819) through a 100nF capacitor (C1). The LED is connected to ground (PWR_FLAGGND). The +12V line also passes through a 10uF capacitor (C3) to the input of a voltage regulator (U1, LM7805_TO220). The regulator's ground pin is connected to PWR_FLAGGND. The regulator's output (+5V) is filtered by a 100nF capacitor (C2) and a 10uF capacitor (C4). The +5V output is connected to an LED (D2) through a 680 ohm resistor (R1). A second +5V output is shown, connected to a +5V input of a second voltage regulator (U3E, LM339) through a 100nF capacitor (C6). The LM339 is also connected to ground (GND) and has a +12V input (U3E V+).

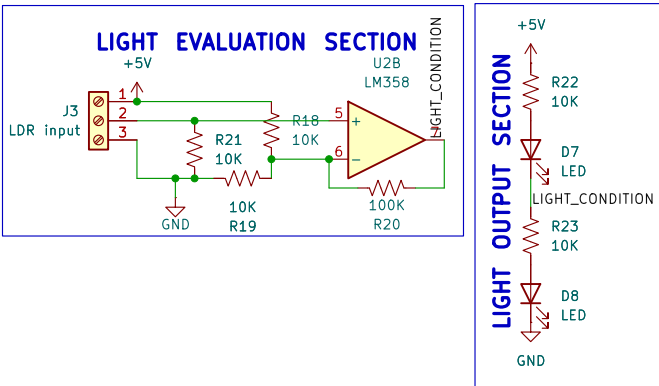


PRESSURE SENSOR SECTION

J4
Pressure input

1 → +5V
2 → R17 10K → PRESSURE - CONDITIONED
3 → GND

1uF C7



PRESSURE EVALUATION SECTION

The diagram illustrates the Pressure Evaluation Section. It features two LM339 comparators, U3D and U3C, configured as voltage comparators. The non-inverting inputs (+) of both comparators are connected to a common node that is pulled up to +5V by resistor R12 (10K) and pulled down to GND by resistor R14 (10K). The inverting inputs (-) are connected to the same node through resistor R13 (10K). The output of U3D is labeled PRES_GT_3.33_N, and the output of U3C is labeled PRES_GT_1.66_N. The comparators are labeled U3D LM339 and U3C LM339.

PRESSURE OUTPUT SECTION

The diagram illustrates the Pressure Output Section, which includes three parallel branches connected to a +5V supply. Each branch contains a resistor (R15, R16, R31) and an LED (D5, D6, D10) in series. The output of the first branch is labeled PRES_GT_3.33_N, and the output of the second branch is labeled PRES_GT_1.66_N. The circuit also includes two BC547 transistors (Q3, Q4) configured as current sources, connected to the +5V supply through resistors (R28, R29) and to the output of the third branch (PRES_GT_1.66_N) through resistors (R30, R31). The emitters of the transistors are connected to GND.

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